

FANUC America Corporation

ZDT GI Data Collector

User's Guide

MAROBZGID11171E REV. A

This manual is to be used with the Cisco® IR809 Data Collector,
Version 1.0.0 and later.

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FANUC America Corporation
3900 W. Hamlin Road
Rochester Hills, Michigan 48309-3253

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

FANUC America Corporation
Training Department
3900 W. Hamlin Road
Rochester Hills, Michigan 48309-3253
www.fanucamerica.com

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One or more of the following U.S. patents might be related to the FANUC products described in this manual.

FANUC America Corporation Patent List

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Conventions



WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.



CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
 - Using a low motion speed, test run the program continuously for at least one full cycle.
 - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

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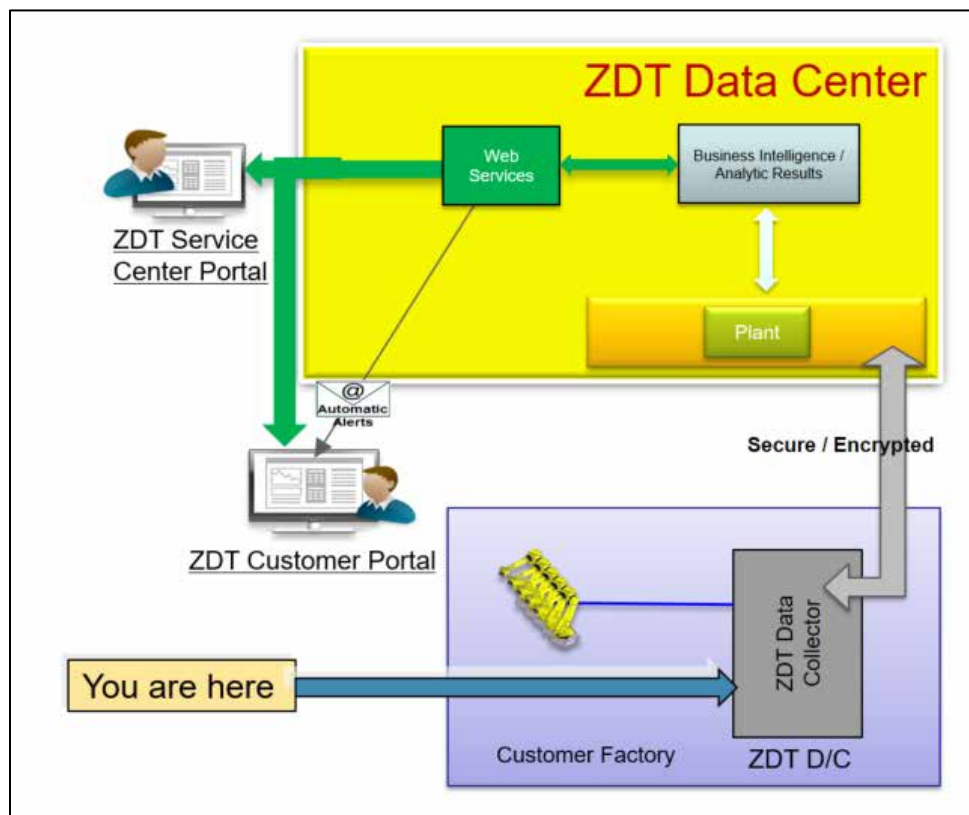
1 Overview

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1 OVERVIEW

The FANUC Zero Down Time (ZDT) GI Data Collector collects and sends out data regarding the performance and state of the connected robots to the FANUC cloud for you to easily monitor information about the devices in your deployment. The GI Data Collector is a hardware appliance that functions as a router and runs FANUC Data Collector software.

Figure 1-1, ZDT Overview



2 Getting Started

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2 GETTING STARTED

Before you can log in to the ZDT GI Data Collector, you must obtain the information and files necessary to complete the network and system configuration. Prepare your network to support the ZDT GI Data Collector.

After these requirements are complete, you can log into the ZDT GI Data Collector to change the default password, install the configuration file and security certificate.

Figure 2-1, FANUC ZDT GI IR809 Data Collector



2.1 Summary of Steps

Complete the following procedures to configure your network and ZDT GI Data Collector. These basic steps must be completed before any ZDT message can be collected and sent to the FANUC data center for analysis and support.

1. *Analyze and thoroughly understand the requirements*
2. *Order FANUC ZDT GI Data Collector*
3. *Cell/Network Configuration*
4. *Power up and Connect IR809 to the Internet*
5. *Contact FANUC to provision the device*
6. *Connect a laptop/PC to the ZDT GI Data Collector*
7. *Log in to the ZDT GI Data Collector*
8. *Install the Site Info file*
9. *Configure the ZDT Client on the robot controllers*
10. *Validate Data Transmission*

2.2 Getting Help

For help and more information, contact support: 1-800-477-6268; press **1** for Robot.

2.3 Requirements

The Zero Down Time (ZDT) GI Data Collector requires the following:

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Table 2-1, Requirements Checklist

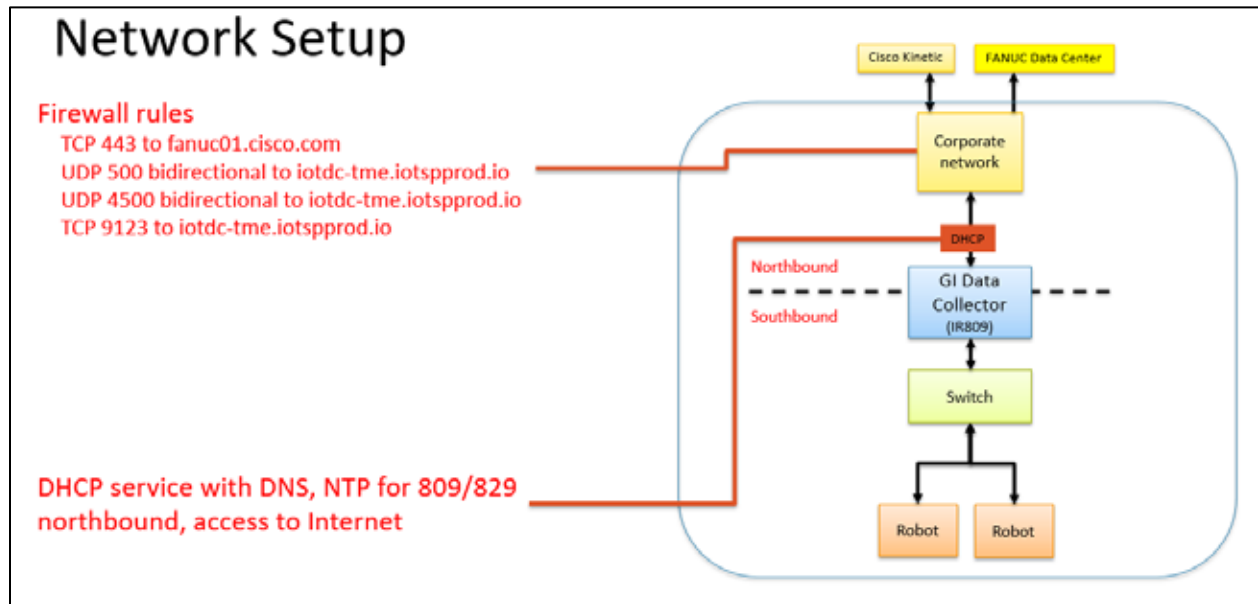
	Requirement	Description	Complete (<i>ü</i>)
IR809	24 V DC power source		☐
	ZDT provisioning file (.zxl)	(Required) ZDT provisioning file will be provided by FAC	☐
	Certificate for HTTPS authentication	(Optional)	☐
Robot	Robot Controller version	V 8.30 or higher	☐
	Robot ZDT software	Preinstalled on R30iB Plus R882 on v 8.30p29 or greater	☐
	Number of robots	GI Data Collector currently supports up to 16 robots	☐
Network	Network architecture	GI Data Collector must function as a network router	☐
	Broadband network		☐
	Network switch	An unmanaged or managed switch if connecting more than one robot	☐
	Ethernet cables		☐
	Customer subnet configuration	Customer specify IR809 router subnet address scheme	☐
	Network information and access	DHCP service with DNS, NTP for 809/829 northbound Firewall rules for Fanuc and Cisco cloud	☐
ZDT portal account	Administrator Corporate site structure Notification users	Administrator of the admin account with a list of corporate site data and users to be notified by ZDT	☐
PC	Supported browser	(Required) Latest version of Chrome (Version 61.0.3163.100) or IE 11. Note If you receive a message that a log or other file cannot be downloaded when using IE 11, install the latest Microsoft security update.	☐
	Supported resolution	(Required) Minimum 1024x768	☐

The GI Data Collector has to be configured as a network router, as indicated in the diagram below.

Data Collector

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Figure 2-2, GI Data Collector Network Architecture



3

Ordering

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3 ORDERING

3.1 Information required to order

The following information is required to order the ZDT GI Data Collector

Table 3-1, Information Required for Ordering

Required Information	Note Entry	Complete (<i>ü</i>)
Information required by FANUC to provide the ZDT provisioning file (.xml): • Customer name • Country • Plant name		☐
Customer IR809 subnet addressing scheme	Specify IR809 southbound subnet IP address scheme. E.g. 192.168.10.1/24	☐

4

Cell/Network Configuration

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4 CELL/NETWORK CONFIGURATION

The following network configuration provides secure web access to the ZDT GI Data Collector web interface. If a managed switch is used to connect robots/PC and IR809, configure the switch to allow traffic between IR809 and the robots/PC. See Figure 2-2.

The following TCP network ports must be open to allow the ZDT GI Data Collector to communicate with the robots and the cloud.

Table 4-1, Required TCP Network Ports

	Ports	Purpose	Open (<i>ü</i>)
Robot Controller to IR809	TCP 6062, 6063	Robots send ZDT messages to ports 6062 and 6063 of the data collector.	☐
IR809 to Robot Controller	TCP 3002	The data collector sends and receives data from robots using port 3002.	☐
IR809 to Robot Controller	TCP 21	Data sent and received via FTP.	☐
PC and IR809	TCP 7443	For accessing Data Collector web interface	☐
IR809 to FANUC data center	TCP 443	The data collector sends secure messages to the cloud using URL <code>fanuc01.cisco.com</code> on port 443.	☐
IR809 from and to Cisco Kinetic	UDP 500, 4500 TCP 9123	Bidirectional UDP to <code>iotdc-tme.iotspprod.io</code> TCP 9123 to <code>iotdc-tme.iotspprod.io</code>	☐

5

Power Up and Connect IR809 to the Internet

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5 POWER UP AND CONNECT IR809 TO THE INTERNET

You may want to write down your IR809 serial number before you setup and mount IR809, as it may be difficult to see the serial number once the unit is mounted. You can find the label as shown below on the back of IR809. Your serial number is in the SN field.

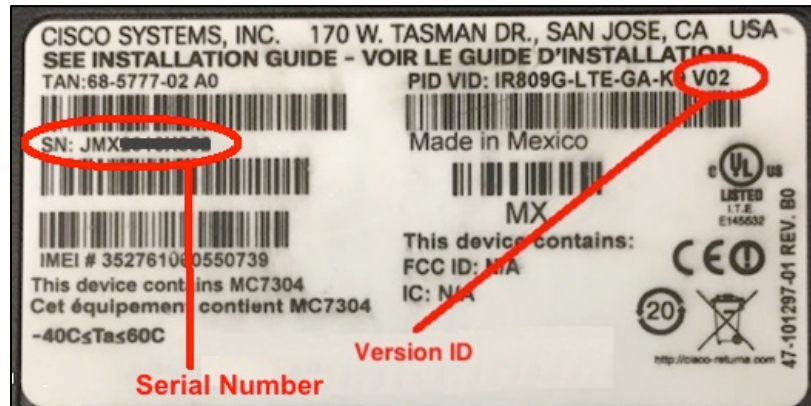


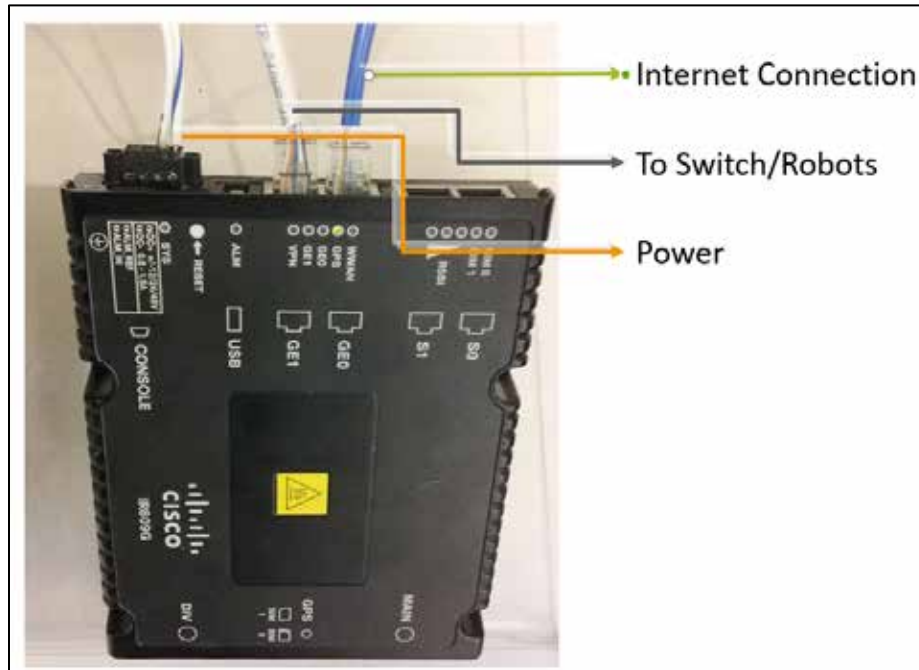
Figure 5.1 Serial number, product ID and version number

Wire the Ethernet cables to the GI Data Collector, robot controllers, network switch, and Internet uplink as shown below.

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Figure 5-1, IR809 Connections



The IR809 contains 2 Gigabit Ethernet ports. GE0, northbound connection, is for communicating with the Internet. GE1, southbound connection, is for the robot subnet. Connect the Ethernet cable with the Internet connection to the IR809 GE0 port. If you have only one robot, connect the GE1 port directly to the robot controller with an Ethernet cable. Otherwise, connect the GE1 port to a network switch, and then connect the robot controllers to this network switch.

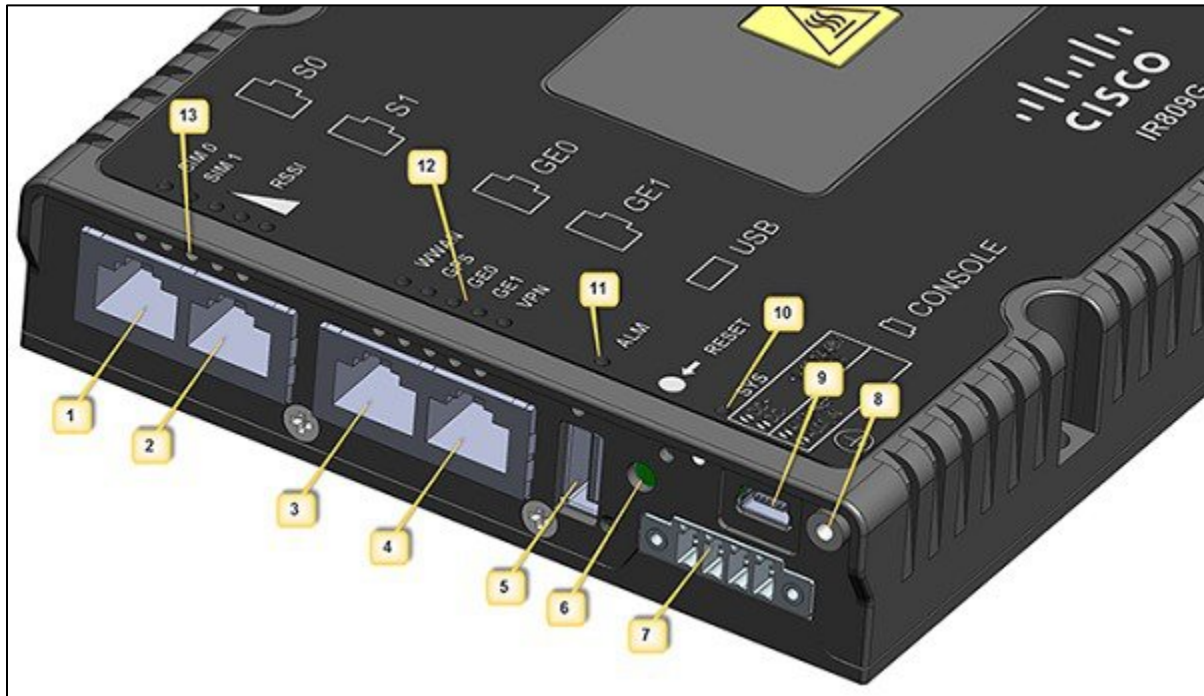
Note: This IR809 setup instruction is a quick setup guide. Please refer to the *Cisco 809 Industrial Integrated Services Router Hardware Installation Guide* as the primary guide for wiring the Cisco IR809. It can be accessed at the following web site.

https://www.cisco.com/c/en/us/td/docs/routers/access/800/809/hardware/install/guide/b_809hwnst/b_809hwinst_chapter_01.html

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Figure 5-2, Cisco IR809



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Table 5-1, Cisco IR809 Features

1	Not used	8	Grounding Point
2	Not used	9	Not used
3	GE0 (10/100/1000)	10	SYS LED
4	GE1 (10/100/1000)	11	Alarm LED
5	Not used	12	WAN/WWAN LEDs
6	RESET Button	13	SIM Card LEDs
7	DC Power/Alarm Connector		

Table 5-2, Cisco IR809 LEDs

LED	Activity	Description
SYS	Power and System Status	Off — No power Green Steady on — Normal operation Green Flashing — Boot up phase or in ROM Monitor mode Amber Steady on — Power is OK but possible internal FPGA program failure
VPN_OK	VPN	Off — No VPN tunnel Steady Green — At least one VPN tunnel is up
GE0 (10/100/1000)WAN 0 GE0 (10/100/1000)WAN 1	Link Status	Off — No link Steady Green — Link is up Flashing — Transmitting and Receiving data

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24 DC is used to power the IR809.

Figure 5-3, DC Power, Alarm Connector

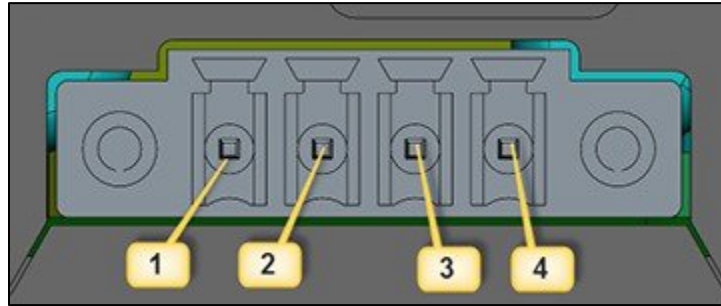


Figure 5-4, DC Alarm, Connector Pins

Pin Number	Name	Description	Color
1	DC In +	DC Power Positive Input	Red
2	DC In -	DC Power Return	Black
3	AC	Alarm Common	N/A
4	AI	Alarm Input	N/A

6

Provisioning the IR809

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6 PROVISIONING THE IR809

The provisioning process registers your product serial number on the Cisco and FANUC portals to allow configuration and communication operations. Please make sure that the information below is available or configured before contacting FANUC cRc to provision the IR809.

IR809 serial number		☐
Robot network configuration	If unknown, FAC will configure IR809 southbound subnet with default setting.	☐
Network information and access	DHCP service with DNS, NTP for 809/829 northbound (Required) Firewall rules applied	☐

Once the information is gathered, you need to submit the information to FANUC to complete the provisioning process.

7 Configuration

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7 CONFIGURATION

7.1.1 ZDT GI Data Collector Configuration

Configuration is the process of setting up the data collector so that the FANUC data center knows that the messages received are for your ZDT subscription. FANUC will provide you a ZDT GI Data Collector web interface URL to enable you to access your data collector once the provisioning process is done.

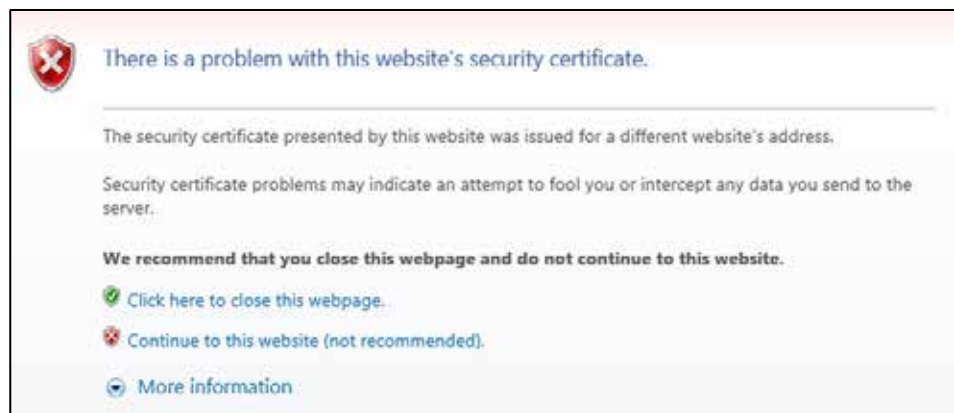
Connect a laptop/PC to the IR809 GE1 port to view the web UI. Make sure that the PC network setting follows the IR809 subnet scheme. i.e. if the IR809 subnet was configured with DHCP, DHCP client should be enabled on the laptop. If no DHCP address was allocated, please manually configure the network on the laptop with a valid IP address, subnet mask, and default gateway IP.

To access the GI Data Collector Web UI use the following procedure.

Procedure 7-1, Accessing the GI Data Collector web UI through a compatible web browser

- Step 1** Open the web browser and enter ZDT GI Data Collector URL, e.g. <https://10.8.188.122:7443>. You will see a warning screen like the one below. **Note:** You will need your IT department to generate and install required certificates to avoid this browser warning message.

Figure 7-1, Security Certificate Warning

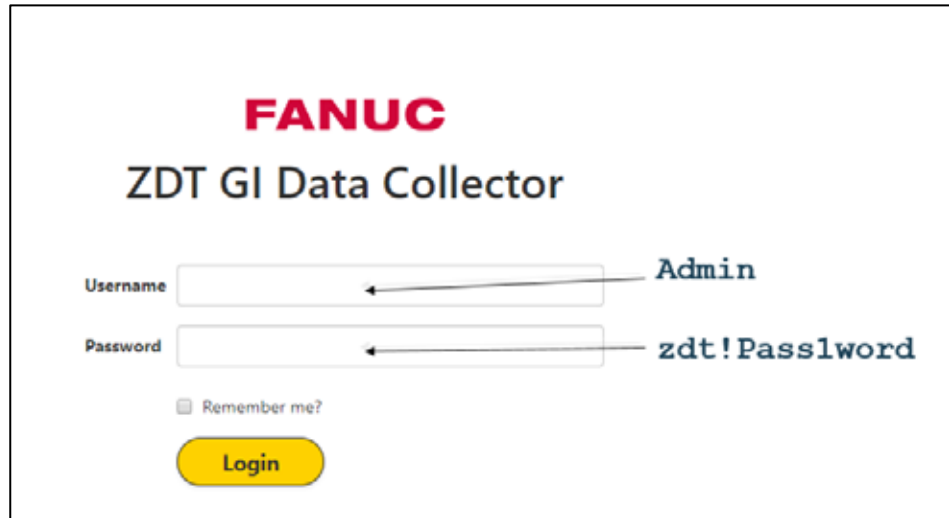


- 2** Click "Continue to this website". This will take you to the GI Data Collector login page as shown in Figure 7-2.
- 3** If you are able to view the Data Collector Web UI, it confirms that the Data Collector is running at the given IP address. Continue to login to ZDT GI Data Collector with default username and password.

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Figure 7-2, Login page



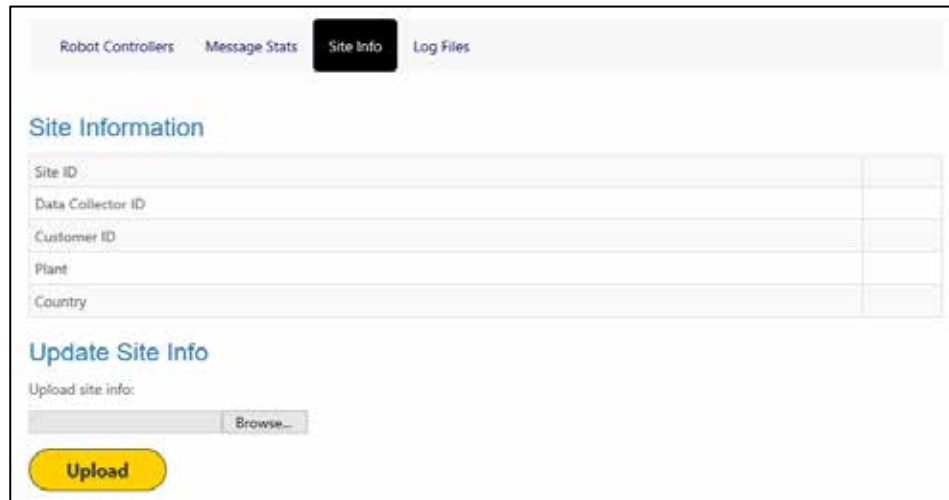
The login page features the FANUC logo at the top, followed by the title 'ZDT GI Data Collector'. Below this are two input fields: 'Username' and 'Password'. Arrows point from the text 'Admin' to the Username field and 'zdt!Pass1word' to the Password field. A 'Remember me?' checkbox is located below the password field. At the bottom is a yellow 'Login' button.

Username: admin

Password: zdt!Pass1word

- 4 Click on the ZDT Data Collector menu, select **Site Info** to install the site info file provided by FANUC. For detail on how to install the Site Info file, please see section 9.3.3, Site Info.

Figure 7-3, Site Info screen



The Site Info screen has a navigation bar at the top with tabs: 'Robot Controllers', 'Message Stats', 'Site Info' (selected), and 'Log Files'. Below the navigation bar is the 'Site Information' section, which contains a table with five rows: 'Site ID', 'Data Collector ID', 'Customer ID', 'Plant', and 'Country'. Each row has a text input field and a corresponding empty cell. Below this table is the 'Update Site Info' section, which includes the text 'Upload site info:', a text input field, a 'Browse...' button, and a yellow 'Upload' button.

7.1.2 ZDT Setup on Robot

You must setup the robot(s) so that they will connect to the IR809 subnet. Robots should be configured with the same subnet address scheme as one another in order to communicate with the ZDT GI Data Collector application.

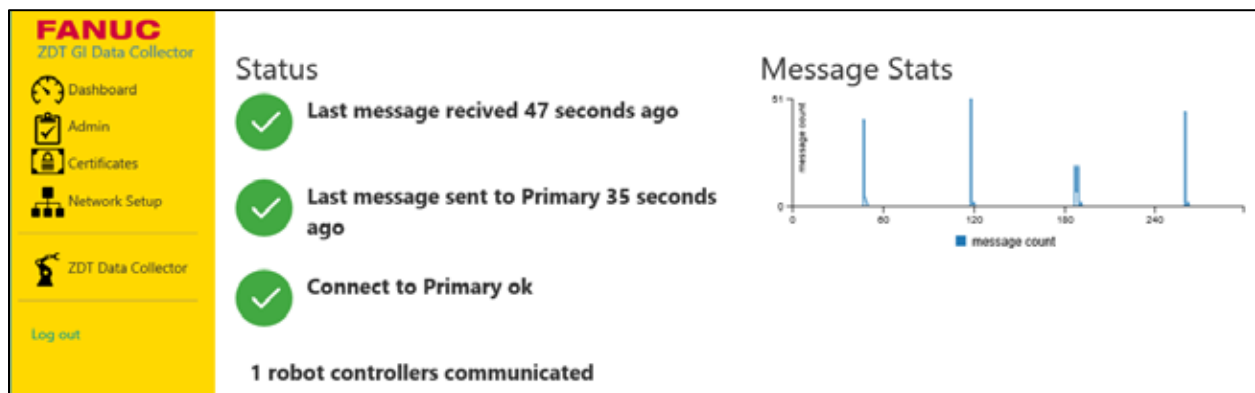
Please refer to the FANUC *R-30iB*, *R-30iB Mate*, and *R-30iB Plus ZDT User's Guide* to configure the robot controllers to send ZDT messages to the GI Data Collector.

8 VALIDATION

8.1 Verify Robot Connections

On the ZDT GI Data Collector Web UI, use the Data Collector Dashboard to check the status. If *Last message received*, *Last message sent*, and *Connect to Primary* are all green, it indicates that the ZDT GI Data Collector is able to receive ZDT messages from the robot controllers and send the messages to the FANUC data center.

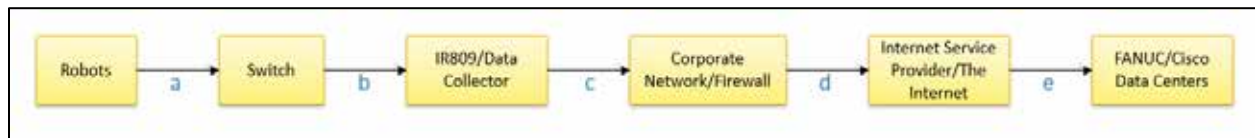
Figure 8-1, Data Collector Dashboard



Go to the ZDT Data Collector, Site Info (see Figure 7-3), and verify that the Site Information table isn't empty. If the Site Information table is empty, please load the FANUC provided site info file.

Below is a chart showing how ZDT messages flow from the robots to the FANUC/Cisco data centers. ZDT messages won't reach the FANUC data center if the communication between any part in the path fails.

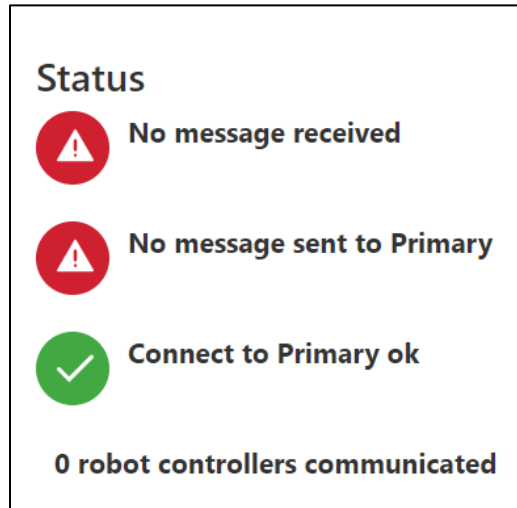
Figure 8-2, ZDT Message Flow



8.1.1 Last message received is red:

The Status indicators found on the Data Collector Dashboard indicate message status.

Figure 8-3, Status (Located on Dashboard)



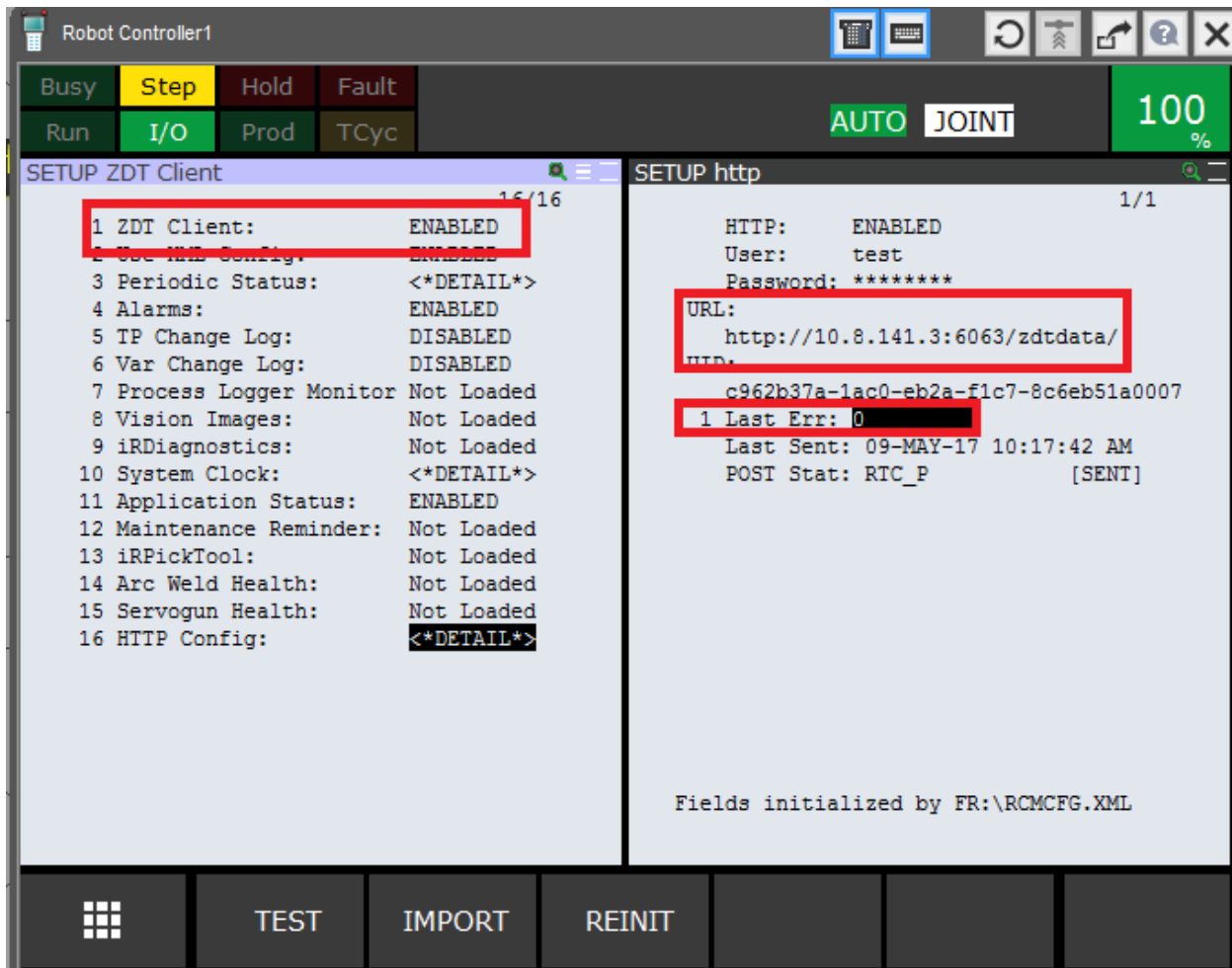
If your robots are unable to send messages to the Data Collector, or messages are blocked during transmission, please go the ZDT Client page on the Teach Pendant and check the settings shown on the following screen.

On the robot teach pendant, go to ZDT Client page:

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Figure 8-4, ZDT Client page on Teach Pendant



In the screen above, make sure:

1. ZDT Client is enabled
16. HTTP Config, URL should have correct ZDT IP address. Last Err should be 0

If you have a managed switch, check to make sure the switch is not blocking the traffic.

8.1.2 Connect to Primary /Last message sent is red:

Please see Dashboard Status, section 9.2.1

8.2 Verify Cloud Transmission

Connect to the FANUC web portal. Contact FANUC to validate the ZDT messages cloud transmission. FANUC will continue the setup of your ZDT web site. You will be able to view your robot health status once FANUC completes your ZDT web site setup.

9

ZDT GI Data Collector

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9 ZDT GI DATA COLLECTOR

9.1 Log in

Log in to the ZDT GI Data Collector and change the default admin password.

Figure 9-1, ZDT GI Data Collector Login



Procedure 9-1, Logging In and Changing the Default Password

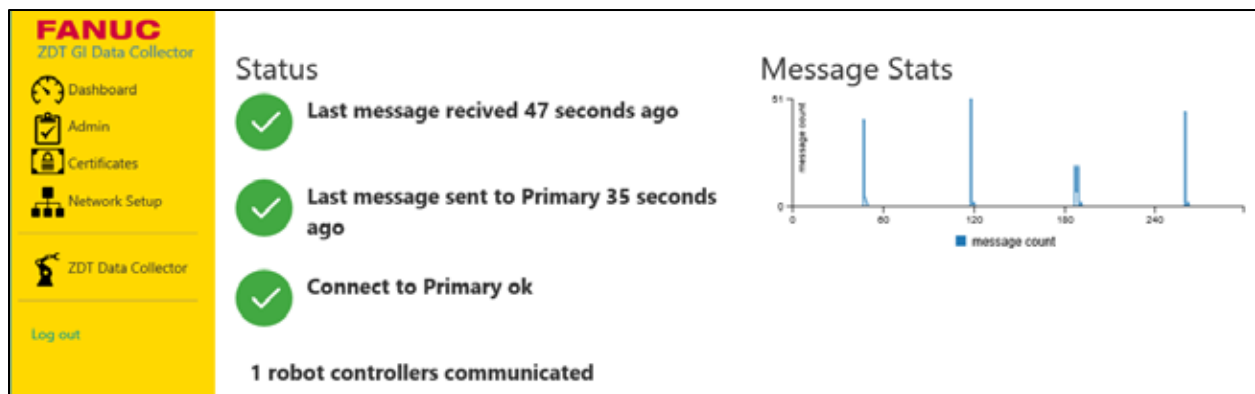
- Step 1** Plug in a laptop/PC to IR809 southbound network. If DHCP is used to configure the southbound network, enabled DHCP client on your PC. If no IP address is available when southbound network is configured, manually enter a valid static IP, subnet mask and gateway address based on the configuration.
- 2** Launch a web browser and enter data collector URL provided by FANUC
- 3** Log in using the default administrator password.
Username: **admin**
Password: **zdt!Pass1word**

- 4 Change the Admin Dashboard Password.
 - a Click **Admin**.
 - b Enter the current admin password.
 - c Enter and re-enter a new password.
 - d Click **Change Password**.

9.2 Dashboard

The Dashboard provides a snapshot of the ZDT GI Data Collector communicating with the associated robots.

Figure 9-2, ZDT Data Collector Dashboard



9.2.1 Status

The Status section provides a high-level snapshot of critical tasks.

Table 9-1, Status

Status	Description
Last Message received	- Green—Messages have been received from robots - Red—No messages have been received from robots for more than 15 minutes. See if ZDT client on robot controller is enabled, and is sending to correct ZDT GI Data Collector IP address.
Last Message Sent	- Green—The last message sent to the primary server succeeded. Communication is successful. - Red—The last message to the primary server failed.

Data Collector

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Status	Description
Connect to Primary	• Green - ZDT GI Data Collector is able to communicate with Data Center. • Red – Unable to communicate with Cisco cloud endpoint fanuc01.cisco.com , check if IR809 northbound has Internet connection and port 443 is not blocked.
X robot controllers communicated	• Indicates how many robot controllers ZDT GI Data Collector has been communicate with.

9.2.2 Message Stats chart

This chart shows the volume of messages received from the robots over time.

9.3 ZDT Data Collector

The ZDT Data Collector dashboard provides dynamically updated information about the system, and the robots that provide data. Click the column headers in each section to sort the data by that category. The following pages are part of the ZDT Data Collector dashboard.

- Robot Controllers
- Message Stats
- Site Info
- Log Files

9.3.1 Robot Controllers

This page includes all robots that are currently sending data to the system. The data is dynamically updated.

Table 9-2, Robot Controllers

Data	Description
Name	The robot name
Controller IP	The robot IP address
Controller ID	The globally unique ID of the robot
Controller Type	The device type. ZDT = R-30iB
Controller State	The current state of the device, such as <i>Enabled</i> or <i>Disabled</i>

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Data	Description
Application	The name of the application tool installed on the robot. For example; SpotTool+
Version	The system software version installed on the robot
Last Updated	When last message received from the robot



Tip Enter a phrase or partial phrase into the Filter to display only robots that include that information. The filter applies to all fields. For example, enter a partial IP address to include devices with that root address.

9.3.2 Message Stats

This page displays how many messages were received from a robot, when a message was last received, how long it took for the message to be forwarded, and other information.

Figure 9-3, Message Stats

Message ID	Controller IP	Name	Line	Cluster	NumReceived	Last Received	NumSent	Avg Send Time	Last Sent
AINFO_T	172.22.200.183	000	ZDITEST	ZZZ	3	2017-11-07T11:05:00	3	0.005	2017-11-07T11:05:11
ALARM_MSG_T	172.22.200.183	000	ZDITEST	ZZZ	20	2017-11-07T10:36:35	20	0.064	2017-11-07T10:36:37
APP_DATA_T	172.22.200.183	000	ZDITEST	ZZZ	3	2017-11-07T11:16:46	3	0.052	2017-11-07T11:16:46
BATT_LOW2_T	172.22.200.183	000	ZDITEST	ZZZ	3	2017-11-07T11:05:00	3	0.004	2017-11-07T11:05:10
CINFO_T	172.22.200.183	000	ZDITEST	ZZZ	1	2017-11-07T11:04:59	1	0.023	2017-11-07T11:05:11
CMEMORY_T	172.22.200.183	000	ZDITEST	ZZZ	3	2017-11-07T11:05:00	3	0.004	2017-11-07T11:05:11
GINFO_T	172.22.200.183	000	ZDITEST	ZZZ	6	2017-11-07T11:04:59	6	0.004	2017-11-07T11:05:12
GJOB_DATA_T	172.22.200.183	000	ZDITEST	ZZZ	75	2017-11-07T11:16:46	75	0.029	2017-11-07T11:16:47
GMAINT_ITM_T	172.22.200.183	000	ZDITEST	ZZZ	9	2017-11-07T10:50:16	9	0.011	2017-11-07T10:50:19
GMAINT_HEAD_T	172.22.200.183	000	ZDITEST	ZZZ	1	2017-11-07T10:50:16	1	0.003	2017-11-07T10:50:16
RTC_T	172.22.200.183	000	ZDITEST	ZZZ	3	2017-11-07T11:05:00	3	0.005	2017-11-07T11:05:12
VIS_RUNDAT_T	172.22.200.183	000	ZDITEST	ZZZ	39	2017-11-07T11:16:46	39	0.188	2017-11-07T11:16:39
VIS_RUNRES_T	172.22.200.183	000	ZDITEST	ZZZ	1651	2017-11-07T11:16:48	1614	0.012	2017-11-07T11:16:50

This information is primarily for display.

Table 9-3, Message States

Data	Description
Message ID	The message name
Controller IP	The robot IP address
Name	The robot name
Line	A site specific name that can be assigned to the robot to help.
Cluster	Cluster is a site specific name that can be assigned to the robot to

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Data	Description
	help identify it.
Num Received	The number of messages received since the last reset
Last Received	When the last message was received
Num Sent	The number of messages sent on
Avg Send Time	The average time for the message to be processed and sent to the primary
Last Sent	When the last message of this type from this robot was sent



Tip Enter a phrase or partial phrase into the Filter to display only robots that include that information. The filter is matched for all fields. For example, enter “VAR” to display only message names that include “VAR”. Enter a partial IP address to include devices with that root address.

9.3.3 Site Info

This page displays data about the ZDT GI Data Collector. Information from this page might be requested by FANUC support personnel when investigating an issue with the GI Data Collector.

Figure 9-4, Site Info

FANUC
ZDT GI Data Collector

 Dashboard

 Admin

 Certificates

 Network Setup

 ZDT Data Collector

Log out

Robot Controllers

Message Stats

Site Info

Log Files

Site Information

Site ID	aee6416d-8fd2-4702-b692-4e7221415c66
Data Collector ID	13978ffd-781b-407b-a9f7-299e2db919fb
Customer ID	fc81aa55-babf-438c-8a44-1bb6542e0dc9
Plant	Venus Lab
Country	USA

Update Site Info

Upload site info:

Data Collector

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Table 9-4, Site Information

Data	Description
Site ID	The globally unique ID of the site the ZDT Data Collector is monitoring
Data Collector ID	The globally unique ID of this ZDT GI Data Collector
Customer ID	The customer's globally unique ID
Plant	Name of the plant
Country	The country where the collector is installed

9.3.4 Log Files

Use Log Files to display logs for the processes that run on the Data Collector (Figure 9-5).

- Select the date of log file to display.
- Use the browser's search tool to search for text within the log output.

Figure 9-5, log Files

The screenshot shows a web interface with a navigation bar at the top containing four tabs: "Robot Controllers", "Message Stats", "Site Info", and "Log Files". The "Log Files" tab is currently selected and highlighted in black. Below the navigation bar, there is a section titled "Select log by date:" followed by a date input field containing "11/07/2017" and a dropdown arrow. Below the date selector is a blue button labeled "Download Logs". The main area of the interface displays a log output with the following text:

```
2017-11-07 15:35:16.667 +00:00 [Information] zdtiox.exe Information: 0 :  
[zdtiox] SiteID          : aee6416d-8fd2-4702-b692-4e7221415c66  
2017-11-07 15:35:17.210 +00:00 [Information] zdtiox.exe Information: 0 :  
[zdtiox] CustomerID     : fc81aa55-babf-438c-8a44-1bb6542e0dc9  
2017-11-07 15:35:17.215 +00:00 [Information] zdtiox.exe Information: 0 :  
[zdtiox] DataCollectorID : 13978ffd-781b-407b-a9f7-299e2db919fb  
2017-11-07 15:35:17.216 +00:00 [Information] zdtiox.exe Information: 0 :  
[zdtiox] Plant          : Venus Lab  
2017-11-07 15:35:18.925 +00:00 [Information] zdtiox.exe Information: 0 :  
[MessageProcessors.StartProcessors] Service started.  
2017-11-07 15:35:18.933 +00:00 [Information] zdtiox.exe Information: 0 :  
[MessageProcessors.StartProcessors] Starting 1 message processing threads.  
2017-11-07 15:35:18.963 +00:00 [Information] zdtiox.exe Information: 0 :  
[MessageStatsAccumulator.Start] Message statistics accumulation started.  
2017-11-07 15:35:18.963 +00:00 [Information] zdtiox.exe Information: 0 :  
[MessageProcessors.OnStart] Service started.  
2017-11-07 15:35:18.974 +00:00 [Information] zdtiox.exe Information: 0 :
```

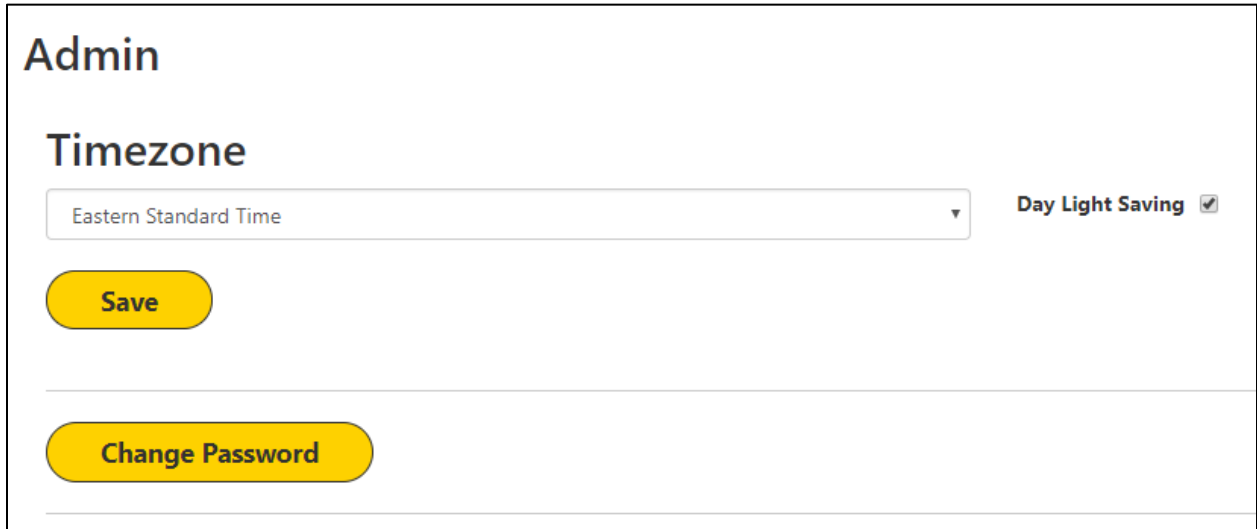


Tip To refresh the log file display, click on another entry and then re-select the file. This prevents the log file from being dynamically updated while you are reviewing it.

9.4 Admin

The admin screen allows you to perform administrative functions.

Figure 9-6, Admin



The screenshot shows the 'Admin' interface. At the top, the word 'Admin' is displayed in a large, bold font. Below it, the 'Timezone' section is highlighted. This section contains a dropdown menu currently set to 'Eastern Standard Time' and a 'Day Light Saving' checkbox which is checked. Below these options are two yellow buttons: 'Save' and 'Change Password'. The 'Change Password' button is located further down the page, separated by a horizontal line from the 'Save' button.

9.4.1 Time zone

Select the time zone; use the Day Light Savings checkbox to enable or disable Day Light Saving, click on Save to Save the change.

9.4.2 Change Password

Click Change Password to change the password.

9.5 Certificates

The following security certificates are required to provide secure communication between the ZDT GI Data Collector, robots, cloud services, and other services.

- **NOTE:** All of the following security certificates are required for proper operation and data reporting on the ZDT GI Data Collector.

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Table 9-5, Required Security Certificates

Security Certificate	Description
HTTPS Server Certificate	ZDT GI Data Collector includes a default HTTPS Server certificate. In order to avoid web browser security warning message, have your IT department generates a web server certificate and add ZDT GI Data Collector IP address to your DNS server. The certificate must be signed using at least SHA256 hashing algorithm.
Satellite-to-Primary Client Certificate	Displays certificate used by GI Data Collector to securely send the messages to the Cisco Cloud.
Trusted Satellite-To-Primary Server Certificate	Displays the Trusted Satellite-To-Primary Server Certificate. This certificate is the public key of the Primary server in the cloud. The satellite uses this certificate to ensure it is talking to a trusted Primary server.

Procedure to Install GI Data Collector Web Server Certificate

Procedure 9-2, Installing Web Server Certificate

Step 1 Click **Certificates**, on the left.

Data Collector

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Replace the HTTPS Web Server Certificate

Figure 9-7, API Certificate Tab

The screenshot shows the FANUC ZDT Gt Data Collector interface. On the left is a yellow sidebar with navigation links: Dashboard, Admin, Certificates, Network Setup, ZDT Data Collector, and Log out. The main content area is titled 'Web Server Certificate' and displays the following details:

Subject DN	CN=zdtgdc2017.fanuczdt.com
Issuer DN	CN=Acme1
Valid From	08/17/2017 13:01:57
Valid To	08/17/2027 03:59:59
Version	3
Signature Algorithm	sha256RSA 1.2.840.113549.1.1.11
Serial Number	3798A114A8C686B44855807F929FCE30
SHA256 Fingerprint	2EE147864472BA6D862704DE7EBDE41B087A14CEC9868A4A9D76769CCCD3D588
Has Private Key	☒

Below the details is the 'Upload certificate' section, which includes a 'Certificate password:' text input field, an 'Upload certificate (pfx format):' text input field with a 'Browse...' button, and a large yellow 'Upload' button at the bottom.

- 2 Upload the certificate
 - a Click the **API** tab.
 - b Click **Choose File** and select the file from a local drive.
 - c Enter certificate password.
 - d Click **Upload** and then **OK**.

10 Troubleshooting

10 TROUBLESHOOTING

10.1 IR809 Subnet and Connectivity to Cisco Kinetic

Connect a PC to the LAN network where the gateway is installed and ensure that the laptop gets a valid IP address. Open DOS Command Prompt:

10.1.1 Verify DHCP

Type in ipconfig to validate the IP address.

10.1.2 Verify DNS

To ensure that DNS is working, connect a laptop or similar device to the LAN network where the gateway is installed and ping a public hostname. E.g.

```
ping www.google.com
```

10.1.3 Verify TPS connectivity

To verify that connectivity to the Tunnel Provisioning Service (TPS) is successful, connect a laptop to the network where the gateway is installed and verify that you can telnet in to your Cisco Kinetic cluster:

```
telnet cluster-address 9123
```

10.2 Why do I get “This site can’t be reached” when I launch the console the first time?

It is possible that the PC you are using is not on the same subnet as the data collector. Try pinging the IP address from the PC. You must use a PC that has access to the GI Data Collector.

10.3 Why are no ZDT messages coming from a specific robot

These are issues you might encounter when managing the connection between the robots and the data collector.

There are several things to check:

- Check the message statistics screen to see if the robot has communicated with the data collector and whether the data has been sent to the FANUC data center.
- The robot controllers must be power cycled when changes are made to the ZDT Client configuration (RCMCFG.XML).

- Verify the “Setup ZDT Client” screen data on the TP.
 - Verify the URL has the correct data collector IP address.
 - The URL is case sensitive so make sure the case of the URL is correct. The data collector expects the following case of the URL ZDTData.

`http://xx.xx.xx.xx:6063/ZDTData.`

10.3.1 Why are the settings in the newly loaded RCMCFG.XML not being used?

You must cycle power on the robot controller for the new settings to take effect.

10.4 Dashboard Status

These are actions you might take when the dashboard indicates a problem.

10.4.1 What if the Message Received is Red?

This is only a problem if you expect that robots are sending messages in the last 15 minutes. For a plant with a large number of robots, you would expect this to always be green. If it is red unexpectedly, there could be several causes. The most likely is that the network connection from the robot to GI Data Collector is down. Try to ping one or more of the robots from IR809 subnet. If none respond, consult with the plant IT department.

10.4.2 What if Connect to Primary/Last message sent is Red?

Check if the corporate firewall is blocking the traffic.

Contact your IT department if authorization through captive portal is required to access the network.

In some rare cases, your Internet Service Provider (ISP) may block certain communication ports. Check if your ISP blocks ports 443 and 9123.

The most likely cause of this is that the link from IR809 to the cloud is down. Connect a laptop to IR809 subnet as in 10.2. Check basic network connectivity by performing an nslookup for fanuc01.cisco.com. If this URL does not resolve to an IP address, then the DNS server is not doing its job. Get your IT department involved.

If fanuc01.cisco.com resolves to an IP address, attempt a telnet connection to fanuc01.cisco.com port 443. If the telnet fails to connect, try the telnet connection from a PC with access to the global Internet. If that responds, the problem is that IR809 has lost connectivity through the firewall. Get your IT department involved. If the PC cannot access fanuc01.cisco.com port 443, call the FANUC Hot-Line.

It is possible but less likely that the trust certificate has expired.

10.4.3 What if the Last Message Sent to Primary is Red?

If Connect to Primary is green, most likely the Internet uplink is too slow. Otherwise, see 10.4.2